

POPULATIONS AND SAMPLES

Introduction

- A population refers to the entire group or collection of individuals, objects, or events that we want to study or make inferences about.
 - All patients with a specific medical condition, all cells in a tissue sample, or all individuals in a certain geographical region.
- A sample is a subset or smaller representative portion of the population that is selected for study or analysis.
 - A group of patients randomly selected from the population of patients with a medical condition, a subset of cells taken from a tissue sample, or a group of individuals randomly chosen from a specific geographical region.

Sampling Terminology

- A **sampling frame** is a list or description of the elements (individuals, objects, events) in the population from which the sample will be selected.
 - Example: A patient registry, a hospital database, or a list of households in a particular area.
- A **sampling unit** is a single element or entity from the sampling frame that can be selected as part of the sample.
 - Example: An individual patient, a specific cell type, or a household.

Sampling Terminology

- **Sampling error** refers to the discrepancy or difference between the characteristics or measurements observed in the sample and the true characteristics or measurements of the population.
 - Example: In biostatistics, sampling error occurs when the characteristics observed in a sample of patients with a medical condition differ from the true characteristics of the entire population of patients with that condition.

Why do we sample?

- Sampling is conducted for several reasons:
- **Practicality**
 - Studying or analyzing an entire population can be time-consuming, expensive, or logistically challenging.
 - Sampling allows researchers to obtain information from a manageable subset of the population, making the study more feasible.

Why do we sample?

- Inference

- Drawing conclusions about the entire population based on a representative sample is possible when the sample accurately reflects the population's characteristics.
- Properly conducted sampling can provide valid inferences about the population.
 - The samples should be unbiased and representative hence generalizability.

Quality and Validity of samples

- Important concepts that relate to the quality and validity of the sample and its ability to accurately reflect the population of interest are:
 - unbiased,
 - representative, and
 - generalizability

Unbiased Samples

- Unbiased sampling refers to a sampling procedure where every element in the population has an equal chance of being selected for the sample.
- In an unbiased sample, there is no systematic favoritism or bias towards certain elements or characteristics in the population.
- Unbiased sampling aims to minimize any potential distortion or skewness in the sample that could arise from a non-random or a biased selection process.

Representative Samples

- A representative sample is one that accurately reflects the characteristics of the population from which it is drawn.
- In a representative sample, the distribution of key variables and characteristics in the sample closely mirrors the distribution in the population.
- An important goal of sampling is to obtain a representative sample to ensure that the findings and conclusions drawn from the sample can be reasonably generalized to the larger population.

Generalizability

- Generalizability refers to the extent to which the findings and conclusions based on a sample can be validly extended or applied to the entire population.
- A sample with high generalizability is one that accurately represents the population and allows for reliable inferences and predictions about the population.
- The more representative and unbiased the sample is, the higher the generalizability of the results.

Summary

- Unbiased sampling ensures that each element in the population has an equal opportunity to be included in the sample.
- A representative sample accurately reflects the characteristics of the population, allowing for valid inferences, while
- Generalizability refers to the extent to which the findings from the sample can be applied to the larger population.
 - By striving for unbiased and representative sampling, researchers can increase the likelihood of obtaining results that are applicable and meaningful to the population of interest.

Probability Sampling

- Probability sampling is a sampling scheme in which every element in the population has a known and non-zero chance of being selected for the sample.
 - **Simple Random Sampling:** Each sample of size n of patients in a population has an equal chance of being selected for the study.
 - **Stratified Sampling:** Patients are divided into different strata (e.g., age groups), and random samples are taken from each stratum.
 - **Systematic Sampling:** Every n^{th} element from a population is selected to form the sample.
 - **Cluster Sampling:** Patients are grouped into clusters (e.g., hospitals), and a random selection of clusters is made, followed by sampling individuals within the selected clusters.

Non-probability Sampling

- Non-probability sampling is a sampling scheme in which the selection of elements from the population is based on subjective judgment or convenience, and the probability of selection is unknown.
 - **Convenience Sampling:** Researchers select patients who are readily available or easily accessible.
 - **Purposive Sampling:** Researchers purposely select individuals who possess specific characteristics or meet certain criteria relevant to the study.
 - **Snowball Sampling:** Researchers identify initial participants who then refer additional participants, creating a snowball effect.

Summary

- NB: while probability sampling allows for statistical inference and generalizability, non-probability sampling is often used in situations where practicality, specific target groups, or exploratory research are the key considerations.
 - Understanding populations, samples, and different sampling schemes is crucial for conducting valid and meaningful biostatistical research.
 - Proper sampling techniques enhance the reliability and generalizability of study findings, leading to more robust conclusions and inferences about the target population.

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Title: Sampling Distributions of Statistics and the Central Limit Theorem

Introduction to Sampling Distributions:

- In statistics, a sampling distribution refers to the distribution of a statistic (e.g., mean, proportion) calculated from multiple samples of the same size drawn from a population.
- It provides insights into the behavior of the statistic and allows us to make inferences about the population based on sample statistics.

Key Concepts:

- **Sample Statistic:** A numerical measure calculated from a sample, such as the sample mean or sample proportion.
- **Population Parameter:** A numerical measure that describes a characteristic of the entire population, such as the population mean or population proportion.
- **Sampling Distribution:** The distribution of sample statistics obtained from multiple random samples drawn from the same population.

Characteristics of Sampling Distributions:

- Sampling distributions tend to have different properties than the population distribution.
- As the sample size increases, the sampling distribution becomes more concentrated around the population parameter.
- Sampling distributions can be used to estimate population parameters and calculate measures of uncertainty, such as standard errors.

Central Limit Theorem (CLT):

- The Central Limit Theorem is a fundamental concept in statistics.
- It states that, under certain conditions, the sampling distribution of the sample mean approximates a normal distribution regardless of the shape of the population distribution.
- The CLT applies to sample means calculated from sufficiently large, independent, and identically distributed (IID) samples.

Key Aspects of the Central Limit Theorem:

- **Sample Size:** As the sample size increases, the sampling distribution of the sample mean becomes increasingly closer to a normal distribution.
- **Independence:** The observations within the sample should be independent of each other.
- **Identically Distributed:** The data should be drawn from the same population with the same distributional characteristics.

Implications of the Central Limit Theorem:

- The CLT allows us to make important inferences about the population mean using sample means and approximate their distribution.
- It provides the basis for constructing confidence intervals and conducting hypothesis tests for population means.

- The CLT is widely applicable, making it valuable in various fields of study.

Practical Applications:

- The Central Limit Theorem is frequently used in hypothesis testing and confidence interval estimation for population means.
- It is particularly useful when the population distribution is unknown or not normally distributed.
- It enables the use of parametric statistical tests and facilitates generalizations from sample results to the population.

Understanding sampling distributions and the Central Limit Theorem is crucial for making inferential conclusions based on sample statistics. These concepts provide a foundation for statistical inference, hypothesis testing, and constructing confidence intervals. The Central Limit Theorem allows for robust and powerful statistical analysis, even when population distribution information is limited or unknown.

Module 5: Sampling & Sampling Distributions

Module Assignment

Source notes cover population/sample terminology, sampling frames, sampling error, probability and non-probability sampling, sampling distributions and the Central Limit Theorem.

1. Research question: What is the prevalence of hypertension among adults attending selected primary-care facilities?
2. Define the target population, study population, sampling frame and sample.
3. Choose and justify an appropriate sampling method.
4. Explain two possible sources of sampling bias and how you would reduce them.
5. Briefly explain how the Central Limit Theorem supports inference from sample data.

Submission guidance: Show working where calculations are required. Interpret statistical results in clinical language. Use Excel, Jamovi or SPSS where appropriate.